Nose Wheel – General Data

1. General

The nose wheel is carried on the axle mounted in the fork of the nose gear, and is of the split hub type to facilitate removal and installation of tubeless tires. An O-ring is installed between the two halves of the hub to provide an airtight housing for the tire. The nose wheel has 6–ply rating 8.90 x 12.50, Type III, low pressure tubeless tire. When Mod S.O.O. 6048, Mod 6/1526 and Mod 6/2183 is embodied, a main hub is fitted with a 36 x 13.00 x 12 tire and tube is installed in the nose wheel position. (Refer to 32-40-62, Nose Wheel (Mod S.O.O. 6048 and Mod 6/2183) – Maintenance Practices).

2. Removal/Installation

Refer to Figure 1 for details.

A. Remove Nose Wheel

1) Jack aircraft until nose wheel is clear of ground. (Refer to 07-00-00, Lifting and Shoring – General Data).

2) Remove axle locating bolts from ends of fork.

3) Remove set screws from adjustable axle spacer and using wrenches SD12552-1 and SD12553-1, loosen off nut of adjustable spacer. Remove lock ring securing fixed spacer to axle.

4) Support nose wheel, remove towing spools, withdraw axle from nose wheel, axle spacers and fork. Lower and remove nose wheel.

5) Remove nose wheel discs from each side of nose wheel.

6) Remove bearing seals and cone bearings from both sides of nose wheel.

7) Clean cone bearings, inspect and repack with grease.

B. Install Nose Wheel

1) Ensure nose wheel bearing housings are clean, then install cone bearings and bearing seals.

2) Install nose wheel discs with plug opposite valve.

3) Lightly lubricate axle with grease (MIL–PRF–23827), support nose wheel in fork and, with locating bolt holes aligned with holes in fork, insert axle through fork bushings and nose wheel together with axle spacers. Ensure that the adjustable axle spacer is installed with the nut adjacent to the fork. Secure fixed spacer to axle with lock ring.

4) Install towing spools and axle locating bolts in each fork end and secure with washers and nuts.

5) Extend adjustable axle spacer, using wrenches SD12552-1 and SD12553-1 until wheel bearings begin to bind, then back off to nearest locking position. Insert set screws to lock adjustable spacer.

6) Check tire for correct inflation. (Refer to 12-10-20, Landing Gear – General Data).

7) Lower aircraft to ground.

Note If installation of nose wheel on aircraft is part of gear configuration change then.

8) Do procedure for airspeed programming instructions as per 32-10-11, Main Gear Leg – General Data, 2.C.

C. Remove Nose Wheel Tire

1) Remove wheel from aircraft.

2) Release all pressure from tire by slackening air valve core then finally remove core completely.

3) Using tire demounters, wooden wedges, or an approved tire machine, separate edges of tire from wheel hub beading.

4) Remove bolts, washers and self-locking nuts, holding hub halves, and remove O-ring and tire. Check O-ring for serviceability and discard if damage.

D. Install Nose Wheel Tire

1) Check hub for scratches, scores and cleanliness. Clean tire area of hub with a dry cleaning solvent. Dust area with french chalk (talc).

2) Position hub half containing air valve on a bench. Place serviceable tire and O-ring on hub, with red spot in tire aligned with air valve.

3) Lubricate threads of bolts and bearing surfaces of nuts, bolt heads, and washers, with an anti-seize compound (MIL–T–5544A).

4) Ensure O-ring is in position and install other half of hub.

Note Ensure that the “L” stamping (which indicates the light side) on the wheel halves are aligned opposite approximately 180 degrees apart.

5) Install bolts, washers and self-locking nuts, and tighten to torque value stamped on wheel.

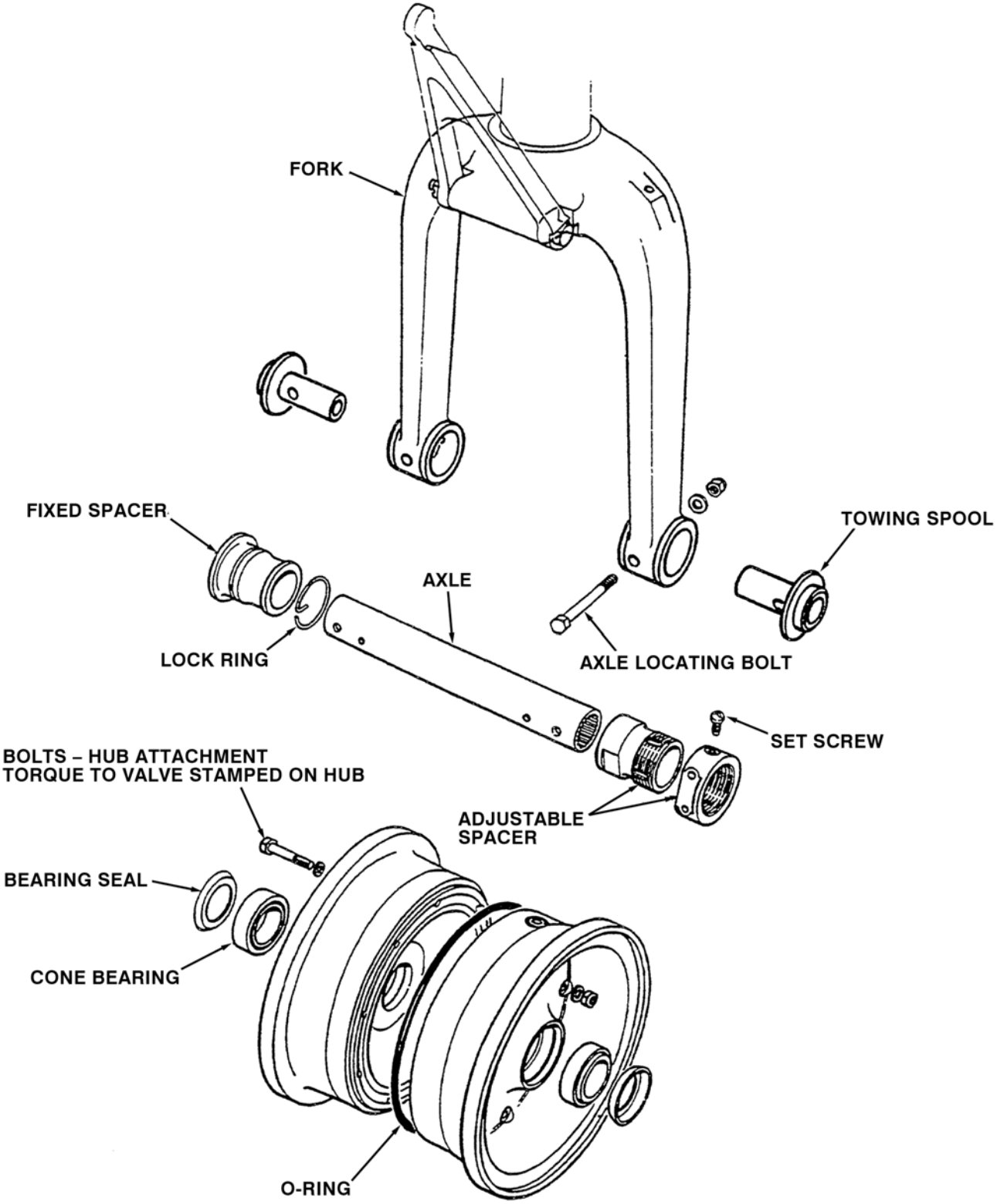
WARNING DO NOT INFLATE TIRE UNTIL ALL BOLTS HAVE BEEN PROPERLY TIGHTENED. IT IS RECOMMENDED THAT ALL MOUNTED TIRES BE INFLATED INSIDE A METAL CAGE TO AVOID POSSIBLE INJURY TO PERSONNEL.

6) Install core in air valve and inflate tire to correct pressure. (Refer to 12-10-20, Landing Gear – General Data).

Note It is recommended that wheel balance is checked when a new tire is installed on hub. The following method of balancing is recommended: check static balance, and add adhesive strip weight as necessary, applying 1/2 of required weight to each side of the hub.

7) Apply a 1” x 3” creep mark using a suitable contrasting coloured paint between tire and rim.

8) Install wheel on aircraft.



Nose Wheel Installation
Figure 1

3. Inspection/Check

A. Inspect Nose Wheel Hub

1) Remove nose wheel and nose wheel tire (refer to Para 2.A and Para 2.C).

2) Clean hub halves with dry cleaning solvent (P–S–661 or equivalent) and dry with compressed air or a lint-free cloth.

3) Inspect hub halves for cracks and breaks using the fluorescent penetrant or an approved dye penetrant method, noting in particular the bead seat, tubewell, and web at junction areas. Replace hub if cracks or chipping of rims is evident.

Note Scratches, nicks and other surface blemishes on hub halves can be removed by sanding with emery cloth removing as little metal as possible. Polish the repaired surface and repaint the repaired areas. Use one coat of Chromated Epoxy Primer to MIL–PRF–23377 on packing groove and mating surfaces of hub halves. Use two coats of Chromated Epoxy Primer to MIL–PRF–23377 and one finish coat of aluminum lacquer (MIL–L–7178) for other areas.

4) Inspect bearing cups and replace if damaged.

5) Install tire to nose wheel and install nose wheel (refer to Para 2.B and Para 2.D).